CellGuide AI

Hackathon plan for cell-context-aware SpCas9 guide selection

Core question

Can an AI agent select different optimal SpCas9 guides for the same target gene depending on the target cell context, by combining sequence-based guide scores with cell-type-specific chromatin accessibility and evidence?

MVP

2 cell contexts

Primary human T cells vs K562; 5-10 genes; experimentally measured indel efficiency.

AI CONTRIBUTION

Multi-objective ranking

Sequence score + ATAC accessibility + off-target risk + evidence/provenance.

WINNING DEMO

Same guide, different context

Show that a sequence-only score is fixed while the cell-aware rank changes with chromatin context.

Executive summary

- Build a reproducible agentic workflow that takes a target gene and cell type, enumerates valid SpCas9 guides, adds cell-specific chromatin features, and ranks the guides for expected editing efficiency and specificity.
- Use the Ito et al. 2024 dataset as the primary hackathon ground truth: 205 gRNAs across 110 genes in primary human T cells, plus a prospective 5-gene validation and explicit T-cell/K562 cross-context tests.
- Treat editing efficiency (indel %) as the primary endpoint. Do not claim that indel percentage is identical to functional knockout probability; add repair/frameshift prediction later.
- Use Claude Code as the orchestrator, Proto for constrained optimization/reranking, Paperclip for literature evidence, Benchling for guide generation/scores when available, and BenchFlow for agent evaluation. ESMC/ESMFold2 are stretch tools for protein-level functional consequence, not guide-efficiency predictors.

Recommended pitch

Existing guide-design tools mostly rank guides from sequence-derived properties. CellGuide AI adds cell-state information and evidence to answer a more specific question: which valid guide is most likely to edit efficiently in this cellular context?

Research-use prototype. The hackathon deliverable should be presented as a computational ranking and benchmarking system, not as a clinically validated guide-design product.

1. Scientific framing and scope

1.1 What is actually cell-context-specific?

The guide sequence itself is determined by the genomic target and PAM. The cell-context-dependent component is the probability that a valid guide can access and edit its genomic target efficiently. Chromatin accessibility, transcriptional state, DNA repair, delivery, and other cellular factors can change observed activity.

Claim	Hackathon interpretation
"Cell-type-specific guide"	A valid genomic guide whose ranking changes when cell-context features change.
Primary endpoint	Measured on-target indel percentage / editing efficiency.
Secondary endpoints	Off-target specificity, rank stability, evidence quality, and uncertainty.
Not yet the endpoint	Protein loss or functional knockout; add repair outcome / frameshift prediction in a later phase.

1.2 Hypotheses

- H1: A sequence-only model provides the same score for a guide in every cell type.
- H2: Adding local ATAC-seq accessibility improves prediction of measured editing efficiency in cell-context-specific datasets.
- H3: For guides evaluated in both T cells and K562, the model can predict the direction of the cell-context difference (higher in T cells vs higher in K562).
- H4 (post-hackathon): A model trained across multiple contexts can generalize to an unseen cell type when given its chromatin profile.

1.3 Keep the MVP deliberately small

Dimension	MVP choice	Reason
Nuclease	Wild-type SpCas9	Largest benchmark ecosystem; avoids mixing nuclease-specific rules.
Edit type	Knockout-oriented cutting / indel formation	Cleanest connection to available ground truth.
Cell contexts	Primary human T cells + K562	Direct published cross-context comparison.
Genes	5-10 for the live demo	Easy to inspect and explain; enough to show context-dependent ranking.
Model	Simple calibrated ranker / regression	A large neural model is unnecessary for a 205-guide core dataset.
Output	Top 3 guides + component scores + evidence	Auditable and demo-friendly.

Important modeling rule

Split or cross-validate by gene, not by random guide, whenever possible. Random guide splits can leak gene/locus properties and overstate generalization.

2. System architecture and tool roles

The agent should not generate arbitrary guide sequences. It should enumerate genomically valid protospacers first, then score and optimize among those candidates.

1	2	3	4	5	6
Input	**Candidates**	**Features**	**Rank**	**Evidence**	**Report**
gene + cell type	valid NGG guides	sequence + ATAC + specificity	multi-objective score	papers + provenance	top guides + uncertainty

Tool	MVP role	Priority	Implementation note
Claude Code	Orchestrate pipeline; write/execute code; call CLI/MCP tools; produce report.	CORE	Use a deterministic Python pipeline underneath; Claude decides which tools to call and explains results.
Proto / proto-tools	Constrained search and reranking over valid guide candidates.	CORE if available	Use propose-score-refine on an enumerated candidate set; do not let it invent non-genomic spacers.
Paperclip	Search/read biomedical literature and supplements; attach evidence to recommendations.	CORE	Evidence layer only; it is not the efficacy predictor.
Benchling	Generate/export candidate gRNAs and on/off-target scores.	CORE / replaceable	If API/export access is limited, substitute CRISPick/CHOPCHOP/local enumeration.
BenchFlow	Evaluate agent task success, tool use, reproducibility, and trajectories.	OPTIONAL MVP	Best used after the biological pipeline works; define fixed benchmark tasks and expected outputs.
ESMC	Protein sequence/function representation for downstream functional prioritization.	STRETCH	Useful for exon/domain consequence; not a gRNA activity model.
ESMFold2	Protein/complex structure analysis for functional consequence.	STRETCH	Use only if targeting a protein domain is part of the objective.

2.1 Minimal local software stack

Python 3.10+; pandas; numpy; scikit-learn; pyBigWig; biopython; pyfaidx; pyranges or bedtools/pybedtools; requests; matplotlib; Git. Add a local CRISPR candidate/scoring tool only if Benchling export is not practical.

Design principle

Keep the biological scoring functions inspectable. The LLM/agent coordinates tools and explains decisions; it should not be the sole numerical predictor.

3. Core hackathon data sources

3.1 Primary ground truth: Ito et al., Nucleic Acids Research (2024)

This study is unusually well matched to the project because it combines experimental SpCas9 editing measurements with chromatin accessibility and explicitly compares cellular contexts.

Component	What is available	How to use it
T-cell development set	205 gRNAs targeting 110 genes; indel percentages measured at least in duplicate.	Benchmark sequence-only scores and fit a small cell-aware model.
Prospective validation	10 newly designed gRNAs across DNMT3A, PDCD1, PRDM1, TGFBR2, MYC.	Use as a small held-out sanity/demo set; do not mix it into model fitting.
Cross-context panel	T-cell-open genes: GZMA, GZMB, CD3D, CD3G, CD28, EOMES. K562-open genes: GATA1, CD33, HBB, HBE1, TFR2.	Primary live demo: test whether cell-aware ranking follows observed T-cell/K562 differences.
T-cell ATAC-seq	GEO GSE221788.	Compute accessibility near each cut site / protospacer.
Independent CAR-T ATAC	GSE231309, sample GSM7256892.	Robustness check: repeat T-cell ATAC scoring with an independent profile.
K562 ATAC-seq	GSE137647, sample GSM4083680.	K562 cell-context features for the same guides.
hMSC-BM ATAC	GSE94272, sample GSM2472047.	Post-hackathon third context / orthogonal validation.
Code	CRISPR-ATAC source on Figshare, DOI 10.6084/m9.figshare.24428509.	Reference implementation for ATAC-aware guide selection.

3.2 Reference genome and annotation

- Use GRCh38/hg38 consistently across guide coordinates, ATAC tracks, and gene models.
- Use GENCODE human gene annotation for transcript/exon/CDS positions; record the release in the final benchmark manifest.
- For every guide, store spacer, PAM, chromosome, cut-site coordinate, strand, target transcript/exon, and genomic build.

3.3 Suggested hackathon 10-gene panel

T-cell-accessible side	K562-accessible side
GZMA	GATA1
GZMB	CD33
CD3D	HBB
CD3G	HBE1
CD28	TFR2

Keep EOMES as an additional T-cell target if time allows. The scientific point is not that a gene "belongs" to one cell type; it is that the tested target sites had contrasting chromatin accessibility in the study.

4. Benchmark design and data schema

4.1 Feature table

Feature block	Example columns	Source
Identity / coordinates	gene, guide_seq, PAM, chr, cut_site, strand, exon	Benchling/local enumerator + GRCh38/GENCODE
Sequence efficacy	DeepSpCas9 / CRISPick / CHOPCHOP score, GC, motifs	Existing scoring tools
Specificity	off-target aggregate score, number of near matches	Benchling or open off-target scorer
Cell context	ATAC signal at cut site; local peak overlap; +/- 50-250 bp window summaries	GEO bigWig/ATAC data
Ground truth	cell_type, measured_indel_pct	Ito supplementary tables
Evidence	paper IDs, cited support, data provenance	Paperclip + curated references

4.2 Models to compare

Model	Inputs	Purpose
Baseline A	Sequence efficacy score only	What a conventional sequence-centric ranker can do.
Baseline B	Sequence + off-target	Separates efficacy/specificity tradeoff.
CellGuide-lite	Sequence + off-target + ATAC	Main hackathon hypothesis test.
CellGuide-agent	CellGuide-lite + literature/provenance + explanation	End-to-end agent demo; biological score stays deterministic.

4.3 Evaluation metrics

- Spearman correlation between predicted rank/score and measured indel percentage.
- MAE or RMSE for calibrated editing-percentage prediction, if you fit a regression model.
- AUROC/AUPRC for "efficient guide" classification; mirror the paper with >50% indel as one interpretable threshold, while also reporting threshold-free metrics.
- Precision@3 / Top-k success: among the top 3 recommended guides for a gene, how often is an experimentally efficient guide recovered?
- Cross-context direction accuracy: for matched guide/target comparisons, does the model correctly predict which cell context has higher editing?
- Gene-held-out validation: group folds by gene or locus to reduce leakage.

4.4 Minimal scoring formulation

CellGuide score

score(g, c) = w_seq * sequence_efficacy(g) + w_atac * accessibility(g, c) + w_spec * specificity(g) + optional functional term. Start with transparent weights or a regularized linear/gradient-boosted model; compare against sequence-only.

For the live demo, emphasize ranking rather than exact probability calibration. The strongest visual is a guide whose sequence score is unchanged but whose ATAC-aware score and observed editing differ between T cells and K562.

5. Hackathon execution plan

Workstream A - Data and benchmark

Step	Output	Done when
A1. Extract Ito supplement	One tidy guide-level CSV with guide, gene, cell context, measured indel %.	Rows reproduce reported counts and gene names.
A2. Collect ATAC tracks	T-cell and K562 bigWig/processed signal files.	Coordinates are hg38 and can be queried programmatically.
A3. Build feature extractor	ATAC signal and peak/window features for each candidate guide.	A single function returns features for any guide + cell type.
A4. Establish baselines	Sequence-only and sequence+ATAC benchmark metrics.	Reproducible results table + plots.

Workstream B - Agent and tools

Step	Output	Done when
B1. Candidate generator	Function/CLI: gene -> valid SpCas9 guides + scores.	Works through Benchling export or an open fallback.
B2. Proto optimizer	Candidate list -> ranked list under efficacy/accessibility/specificity constraints.	Never outputs a spacer absent from the enumerated genomic set.
B3. Paperclip evidence	Target/gene/cell query -> compact evidence bundle.	Every recommendation can show the supporting source and data origin.
B4. Claude Code orchestration	One command/prompt executes data retrieval, scoring, ranking, and report.	A fresh target runs end-to-end without manual notebook editing.

Workstream C - Demo

- Screen 1: choose cell type and target gene.
- Screen 2: show candidate guides with sequence, off-target, and ATAC components.
- Screen 3: compare the same target in T cells vs K562; highlight rank reversals or strong score shifts.
- Screen 4: show experimental indel ground truth and benchmark metric.
- Screen 5: "Why this guide?" evidence/provenance generated from Paperclip plus deterministic feature values.

Minimum viable success

Even if the full agent UI is unfinished, the project succeeds if you can reproducibly show: (1) valid guide enumeration, (2) cell-specific ATAC features, (3) improved or complementary ranking vs a sequence-only baseline, and (4) experimental validation on the T-cell/K562 panel.

6. Post-hackathon expansion roadmap

Expand in tiers based on the quality of the biological ground truth. Direct editing measurements are more appropriate than viability/depletion screens for a model that claims to predict guide editing efficiency.

Tier	Dataset/context	What it adds	Caveat
1	Ito hMSC-BM + public ATAC (GSE94272/GSM2472047)	A third cell context using the same chromatin-aware design logic.	Smaller validation than the T-cell development set.
1	Xu et al. 2018: HEK293FT, human iPSC, primary MSC, primary T cells; B2M/APP/AAVS1/OCT4/PDCD1	Direct RNP editing across very different cell types; useful negative/positive test of context dependence.	Small number of loci; T7EN1-based efficiency in many comparisons; delivery protocol is a major factor.
1	Jensen et al. 2017: 20 gRNAs / 10 loci in HEK293T, with HeLa validation	Orthogonal evidence that chromatin accessibility affects Cas9 editing.	Small dataset; not a broad multi-cell benchmark.
2	Leenay et al. / SPROUT: 1,656 endogenous sites in primary human T cells	Repair outcome, indel length/probability, frameshift-related features; moves toward functional KO probability.	Single main cellular context; repair prediction is a different endpoint from cleavage/editing.
2	DeepHF: 55,604 WT-SpCas9 guide indel rates in HEK293T integrated targets + 85 endogenous WT sites	Large sequence-learning resource and pretrained baseline.	Most large-screen labels are integrated/synthetic target activity, not cell-context-specific endogenous editing.
3	Genome-scale knockout/DepMap-style screens across many cell lines	Large diversity for downstream functional phenotype and essentiality.	Do not treat depletion as direct editing efficiency; gene essentiality and growth strongly confound the label.

6.1 Model evolution

Stage	Model question
Hackathon	Does ATAC information improve ranking within T cells and explain T-cell/K562 differences?
v2 multi-context	Can one model predict editing from guide + locus + cell-context features across several cell types?
v3 unseen-cell transfer	Can the model generalize to a cell type absent during training using only its chromatin/transcriptome profile?
v4 functional KO	Can editing probability be combined with repair outcome and exon/domain importance to predict functional knockout?

6.2 Where ESMC / ESMFold2 fit later

For a knockout-oriented extension, protein models can help prioritize which exon or protein region is most likely to produce functional disruption. This is a downstream objective: first predict whether the genomic cut will occur; then estimate whether the resulting edit is likely to disrupt protein function.

7. Risks, decisions, and final deliverables

Risk	Mitigation
Chromatin is not the only cell-specific determinant.	Model ATAC as one feature; explicitly discuss delivery, repair state, transcription, and assay differences.
Small cross-cell benchmark.	Use it as a mechanistic validation, not a large training set. Expand with orthogonal direct-editing datasets later.
Indel % is not functional KO.	Name the endpoint correctly; add SPROUT/frameshift and exon/domain consequence in v2.
Guide-score/API access varies across tools.	Make the pipeline modular; Benchling can be replaced by open guide enumeration/scoring.
Agent hallucination or unreproducible recommendations.	Numerical scores come from deterministic functions; agent output includes provenance and raw component scores.
Leakage across guides from the same gene/locus.	Use gene/locus-held-out evaluation and clearly separate prospective validation guides.

7.1 Deliverables

- A reproducible repository with environment setup, data manifest, and one-command benchmark.
- guide_benchmark.csv: harmonized guide-level labels and cell-context features.
- A baseline results figure: sequence-only vs sequence+ATAC.
- A T-cell vs K562 comparison for 5-10 genes, with experimental ground truth.
- An agent demo that returns the top 3 guides, component scores, evidence, and uncertainty/limitations.
- A short README describing what is experimentally validated versus inferred.

7.2 Go / no-go criteria for the weekend

By milestone	Go criterion
After data ingestion	Experimental guide labels and T/K562 ATAC features map to the same hg38 coordinates.
After baseline	At least one transparent cell-aware analysis can be compared quantitatively with sequence-only.
Before UI polish	The pipeline produces a stable ranked table for a target gene in each cell context.
Final demo	A judge can see why the recommended guide differs by cellular context and can trace every score to data/source.

Bottom line

This is a strong hackathon project because it has a narrow, falsifiable hypothesis and public experimental ground truth. The best scope is not "AI invents CRISPR guides," but "an auditable AI system reranks valid guides using cellular context and shows whether that improves experimentally validated selection."

8. Key sources and links

1. Ito Y. et al. Epigenetic profiles guide improved CRISPR/Cas9-mediated gene editing in human T cells. Nucleic Acids Research 52(1):141-154 (2024). [link]
2. CRISPR-ATAC source code (Ito/Kagoya group), Figshare DOI 10.6084/m9.figshare.24428509. [link]
3. GEO GSE221788 - T-cell ATAC-seq used by Ito et al. [link]
4. GEO GSE137647 / GSM4083680 - K562 ATAC-seq used by Ito et al. [link]
5. GEO GSE231309 / GSM7256892 - independent CD8 CAR-T ATAC-seq used for validation. [link]
6. GEO GSE94272 / GSM2472047 - human bone-marrow MSC ATAC-seq used by Ito et al. [link]
7. Anthropic: Connect Claude Code to tools via MCP. [link]
8. Proto Language - constraint-based biological sequence design with propose-score-refine. [link]
9. Paperclip - CLI/MCP server for biomedical literature, trials, and regulatory documents. [link]
10. Benchling CRISPR guide RNA design - candidate design plus on/off-target scoring. [link]
11. BenchFlow - agent evaluation runtime and benchmark environments. [link]
12. Biohub: ESMC and ESMFold2 protein modeling/structure tools. [link]
13. Xu X. et al. Efficient homology-directed gene editing by CRISPR/Cas9 in human stem and primary cells using tube electroporation. Scientific Reports 8:11649 (2018). [link]
14. Leenay R.T. et al. Large dataset enables prediction of repair after CRISPR-Cas9 editing in primary T cells. Nature Biotechnology 37:1034-1037 (2019). [link]
15. Wang D. et al. Optimized CRISPR guide RNA design for two high-fidelity Cas9 variants by deep learning (DeepHF). Nature Communications 10:4284 (2019). [link]
16. Jensen K.T. et al. Chromatin accessibility and guide sequence secondary structure affect CRISPR-Cas9 gene editing efficiency. FEBS Letters (2017). [link]
17. GENCODE Human - reference gene annotation. [link]

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